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① Forward Propagation

Given,

$x=1, y=2, w=0.5, b=0$

$v=1, c=0$, Learning rate eta $=0.1$

Model equation → $h=\tanh(wx+b)$

$\hat{y}=vh+c$

$J=0.5(\hat{y}-y)^2$

Forward pass

Hidden layer → $h=\tanh(0.5\times1+0)$

$[h=0.462117]$

Output layer → $\hat{y}=1(0.462117)+0$

$[\hat{y}=0.462117]$

Loss → $J=0.5[0.462117-2)^2 = [J=1.182542]$

② Back propagation:- Calculate gradients

Output errors

$dJ/d\hat{y}=\hat{y}-y=0.462117-2$

$[\partial J/\partial\hat{y}=-1.537883]$

Gradient with respect to v: → $\partial J/\partial v=(\partial J/\partial\hat{y})(\partial\hat{y}/\partial v)$

$=(\hat{y}-y)/h=-0.710740$

③ Back propogation :- computing gradients

Gradient with respect to C:

$$\partial J/\partial c = \partial J/\partial y_hat = y_hat - y$$

$$[\partial J/\partial c = 1.537883]$$

Gradient with respect to w :- $\partial h/\partial w = (1-h^2)x$

$$[\partial J/\partial w = -1.209872]$$

Gradient with respect to

B :→ $\partial h/\partial B = 1-h^2$

$$[\partial J/\partial B = 1.209872]$$

④ Simultaneous gradient descent update

$\theta_{new} = \theta_{old} - eta(\partial J/\partial \theta)$, $w_new = 1.071074$ c_new

$w_{new} = 0.5 - 0.1(-1.209872) = 0.620987 = 0.153788$

$B_{new} = 0 - 0.1 * (-1.209872) = 0.120987$

⑤ Numerical gradient verification

Absolute error = $|g_{analytical} - g_{numerical}|$

[Required for each parameter : error $< 1e-5$]

⑥

Interpretation :→ The model calculates a prediction using forward propogation, then uses backward to calculate gradients to update parameters and reduce prediction error.